isonavi-rov

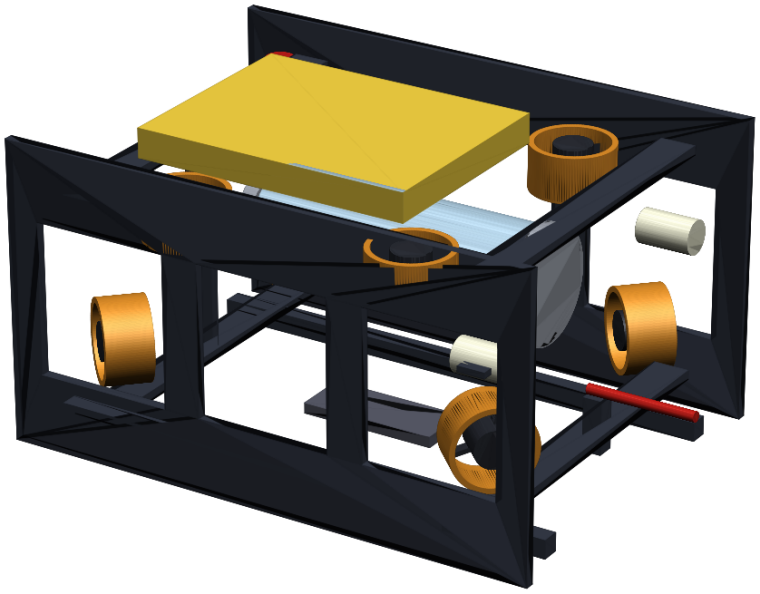
A tethered, piloted underwater vehicle for confined water

Technical Design Report

Vehicle, power, safety, vision and pilot assist, and the simulation that sizes them

Summary. This report covers the simulation study of isonavi-rov on its parametric reference vehicle: a 10.48 kg, 66 by 44 by 33 cm open frame ROV, powered and controlled over a 50 m tether, that threads a 1 m gate, finds and touches a coloured target in an unlit pool by its own light, and flies a three section walled maze. The reference carries eight vectored thrusters; the build carries six on six ESCs, and the layout is a parameter of the budget, so every result here reruns on it.

The design is closed by analysis rather than chosen: mass, buoyancy and trim are solved from a component list and checked against a CAD model built from the same list; the power path is sized to a governed peak; and the thrusters are matched to their propeller. Four pilot modes, manual, assisted, the camera alone in the lateral loop, and shared, where the camera's correction and the pilot's stick are blended by the error each has, are flown through the course in simulation with a modelled human operator, across water clarities and forty random course layouts, and scored on contacts, target and time. The vehicle-side software then runs on a hardware-in-the-loop bench over a framed, CRC-checked protocol, and reproduces the simulation.



Mass, envelope	10.48 kg, 66 x 44 x 33 cm	Contacts per run, murky: assist, shared	5.85, 0.45
Surge force, one unit lost	42.4 N	Contacts per run, turbid: assist, shared	0.45, 0.40
Governed input, peak current	800 W, 16.6 A	Target colour, 1 m, murky water	100 %
Link loss to zero thrust	0.52 s	Cost, vehicle with tether and surface	Rs. 1,15,079

Every number in this report is generated from the result files
by a script in the repository that accompanies it

`github.com/AUV-VITP/isonavi-rov`

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1 Requirements

The vehicle is designed against the following, which together describe a piloted pool task: pass a submerged gate without touching it, identify a target by colour in an unlit pool and touch it, and fly a walled maze while touching as little as possible.

Requirement	Value	How it is met
Mass in air	below 35 kg, target below 20 kg	10.48 kg, Section 2
Envelope	target below 75 x 65 x 60 cm	66 x 44 x 33 cm over the probe
Power	230 V 50 Hz mains, fused	54 V supply, 25 A fuse, Section 3
Emergency stop	external kill switch	four layer chain, Section 4
Tether	about 50 m	power pair and Cat5e, neutrally buoyant
Pool	20 x 10 m, 0.6 to 1.8 m deep	arena model, Section 5
Gate	1 m clear opening	beam 44 cm
Maze walls	up to 1.5 m high, covered passage	arena model
Leaks and pollution	none	no battery aboard, sealed hull, leak probes
Compressed gas, lasers	none carried	not applicable

Table 1: Design requirements and where each is closed.

2 Vehicle

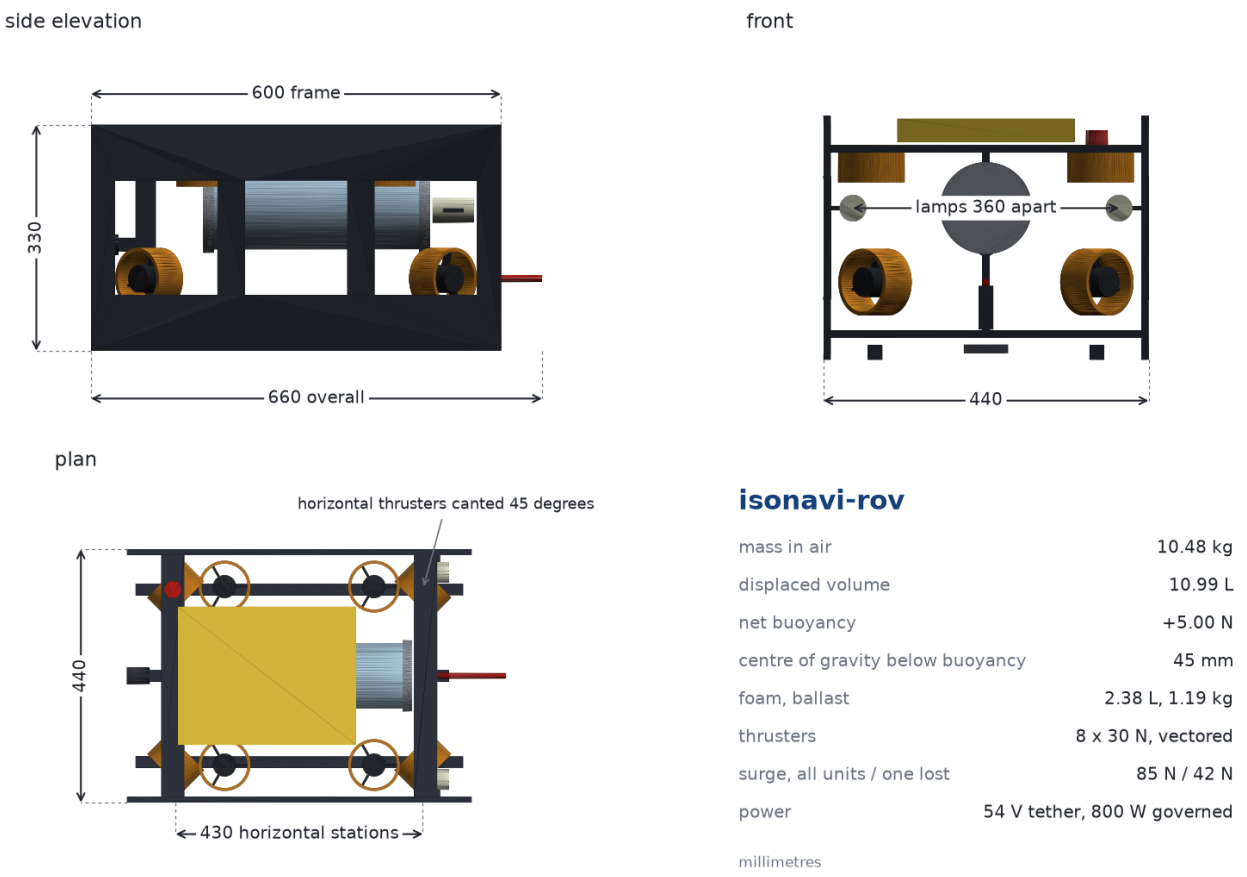


Figure 1: General arrangement, from the CAD model built out of the solved budget. Four horizontal thrusters canted at 45 degrees give surge, sway and yaw; four vertical thrusters give heave, roll and pitch. The vehicle can move sideways without turning, which is how it crosses the slalom without widening its swept path.

2.1 Mass, buoyancy and trim are solved

Every component is listed with a mass, a displaced volume and a station (`simulation/vehicle/rov.py`). Three things are then solved rather than chosen: the foam volume, so the vehicle floats 5.00 N positive and rises if power is lost; the lead ballast, so the centre of gravity sits 45 mm below the centre of buoyancy and the vehicle rights itself; and the foam station, so it floats level (trim offset 0.00 mm). The result is 10.48 kg and 10.99 L with 2.38 L of foam and 1.19 kg of lead.

2.2 The CAD checks the budget

`simulation/vehicle/rov_cad.py` builds the 39-part assembly from the solved budget and checks it three ways. The displaced volume of each group as built agrees with the budget: buoyancy 2.486 L and hull 3.449 L exactly, the rest to within items not modelled as solids (fasteners, cables, the depth sensor). The assembly fits a 66.0 x 44.0 x 33.0 cm box. And every part is tested against every other for overlap: 0 pairs overlap, the closest thruster duct clears a side plate by 10.0 mm, and the foam blocks 4 mL of the worst vertical thruster's outflow.

The check earned its place. The first build found the vertical ducts touching the side plates, which moved their station inboard by 15 mm; displaced volumes for the ducts, skids and end caps that the budget had underestimated; a probe that protrudes 6 cm past the frame and therefore had to be in the contact model; lamps that overlapped the front horizontal ducts; and a kill switch inside the foam. Each was fixed in the budget, not only in the drawing, and the simulation re-run.

2.3 Hydrodynamics and thrust

Damping and added mass are scaled from the tank-identified coefficients of the BlueROV2 Heavy, an open frame vehicle of the same class, by projected area and displaced volume. With 30 N thrusters the vehicle makes 84.9 N in surge and sway and 118.2 N in heave, for 0.73 $\mathrm{m\,s^{-1}}$ ahead.

The course is flown at 0.35 $\mathrm{m\,s^{-1}}$. There, the tether is a third of the load: 8 m of trailing tether adds 9.4 N to the frame's 21.9 N. With any one thruster lost, a linear program over the surviving seven still makes 42.4 N with zero net moment, a margin of 1.4 over that load. That margin is what set the thrust target at 30 N.

The tether drag acts where the tether is terminated, so the termination sits at the height of the centre of gravity, on a post between the two rear cross members: 3 mm from the CG height, the drag at operating speed holds 0.3° of pitch against the righting moment of the BG, which the attitude hold barely has to answer.

The horizontal thrusters sit 53 mm below the CG, so surge thrust pitches the vehicle: at the full surge envelope that is 4.5 N m against a righting moment that peaks at $mg\,BG$, and uncorrected it would hold the vehicle about 76 degrees nose up. The allocation therefore takes every moment about the CG with the thrusters at their real heights, and answers surge with the verticals; the envelope above already pays for it. The python dynamics had left the heights out until the Gazebo model (Section 10), which places the thrusters where the CAD has them, made the coupling visible.

2.4 Thrusters, made in house

Each thruster is a hobby outrunner run flooded, with a potted stator, in a printed Kort duct with a 76 mm three blade propeller. The motor is matched to the propeller rather than picked by habit (`simulation/vehicle/thruster.py`): at the bench point the propeller turns at about 2845 rpm, far below where a high KV winding wants to run, so the motor works at low speed and high torque. A 1000 KV winding would carry 27.4 A of phase current, past 80 % of a 30 A controller. The 750 KV winding carries 20.5 A and draws 114 W, and its motor and propeller efficiencies together reproduce the drive efficiency the power path was sized on. The propellers are printed in left and right hand pairs so their reaction torques cancel; the dynamics leave propeller torque out, which holds only with them paired.

3 Power and tether

A 48 V mains supply at the poolside is set to 54 V, inside its 43 to 55 V adjustment and below its over-voltage threshold, and that runs down two paralleled 2.5 $\mathrm{mm^2}$ cores per leg (loop resistance 0.344 Ω) to two converters aboard that feed the thruster rail. The higher voltage carries the same power on less current, and tether loss goes with the square of the current. There is no battery on the vehicle.

All eight thrusters at full draw 1079 W, which would need a 40 A fuse and a supply to match. That is not a design point, so a governor in the allocator caps input at 800 W. Propeller power from momentum theory is homogeneous of degree 3/2 in thrust, so scaling every thruster by k scales the total by $k^{3/2}$ and the k that meets the cap is exact in one step. At the cap all eight together still make 24.3 N each, the current is 16.6 A, the converters see 48.3 V against a 36 V floor, and the tether dissipates 94 W. The surface supply is 1000 W and the main fuse 25 A, at 1.25 times the governed peak current. Fifty metres is inside the 100 m reach of plain Ethernet, so the data pair is Cat5e with no modem.

At the governed peak the converters lose 48 W as heat inside the sealed acrylic tube, where still air and the acrylic wall both conduct poorly. An aluminium plate carries it from the converters to the rear end cap, which is machined aluminium and in the water.

4 Safety chain

Four layers, each able to stop the thrusters on its own:

1. a fused mains supply with an emergency stop on its output;
2. an external magnet kill switch on the vehicle that opens the thruster rail;
3. the supervisor on the flight computer, which zeroes thrust on a kill, a leak, or 0.5 s without a pilot packet, and latches the kill and leak states;
4. the ESP32, whose firmware drops every channel to neutral if the flight computer goes quiet.

The vehicle floats positive, so zero thrust is the safe state: from the deepest point of the pool it rises unpowered. On the hardware-in-the-loop bench (Section 9) a 10 s loss of pilot packets put the board in HOLD 0.52 s after the last packet, one control tick after the timeout, with zero thrust throughout and a rise of 0.87 m; packets returning re-armed it. A kill latched on the tick it arrived and held zero thrust for all 499 ticks that followed, through continuing pilot traffic.

5 Arena and camera

5.1 Arena

The pool is 20 m by 10 m, its floor sloping from 0.6 m to 1.8 m. The course is a gate of two posts and a crossbar with a 1 m opening; four balloons of different colours on a line, one of which is the target; and a three section maze, a straight corridor, a slalom between posts, and a covered passage, with walls up to 1.5 m high. Placement is parametric, and the results are reported both for a reference layout and for forty layouts drawn at random. Each element carries a signed distance function, and the hull and probe are sampled as points, so a contact is any sample on or inside an element.

5.2 Camera image formation

In an unlit pool the only light is what the vehicle carries. Per pixel and per colour channel, with c the attenuation coefficient, β the volume backscatter coefficient, ρ the surface albedo, t the range along the ray and r_l the lamp to point range,

$$L = \frac{\rho}{\pi} \sum_{\text{lamps}} I_0 \, b(\theta) \cos i \, \frac{e^{-c r_l}}{r_l^2} \, e^{-c t}, \qquad B = \int_0^t \beta \sum_{\text{lamps}} I_0 \, b(\theta_s) \, \frac{e^{-c r_l(s)}}{r_l(s)^2} \, e^{-c s} \, ds.$$

Red is absorbed several times faster than blue, so colour fails before detection does, and the backscatter integral is evaluated along every ray, so lamp placement shows in the image the way it does in a real one. Water clarity is set by the scattering coefficient b: 0.05 per metre for a clear pool, 0.4 for a turbid one and 1.0 for murky water.

6 Vision

6.1 Target colour, without training data

A fixed colour threshold works at one range in one water and nowhere else, so the target finder undoes the water instead. It removes the backscatter veil, segments what remains into round blobs, takes range from apparent size against the known target radius, multiplies each channel by e^{2cr} to undo the two way attenuation at that range, and matches the corrected chromaticity to the nearest reference. A blob whose

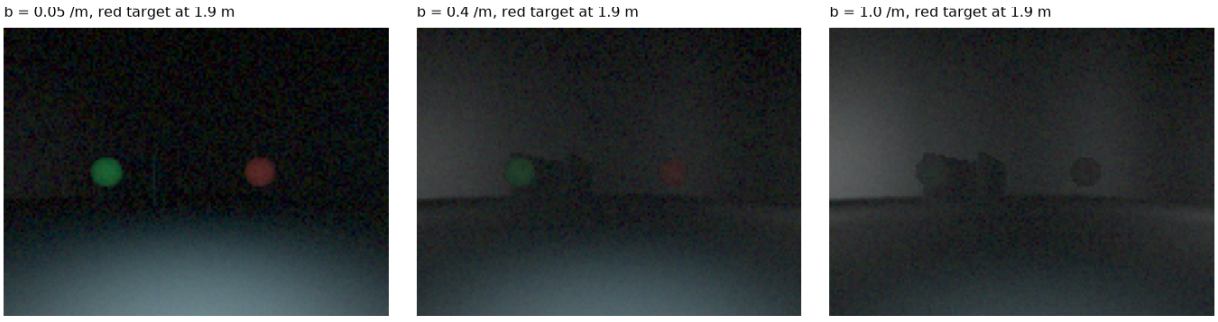


Figure 2: The onboard camera at the same pose in three water clarities, rendered by the camera model. In murky water the balloons are still there; their colour is not.

size implies a range the lamps cannot light is speckle and is rejected; left in, the correction at that range turns it any colour.

With the lamps as built, 0.18 m either side of the lens, the target colour is identified at 1 m in 100, 100 and 100 % of frames in clear, turbid and murky water, and at 1.5 m in 100, 78 and 25 %. Lamp placement matters: at 1.5 m in turbid water, lamps 0.10 m off the lens identify 41 % and lamps as built 78 %. The operating rule that follows is to close to about a metre before committing to a colour.

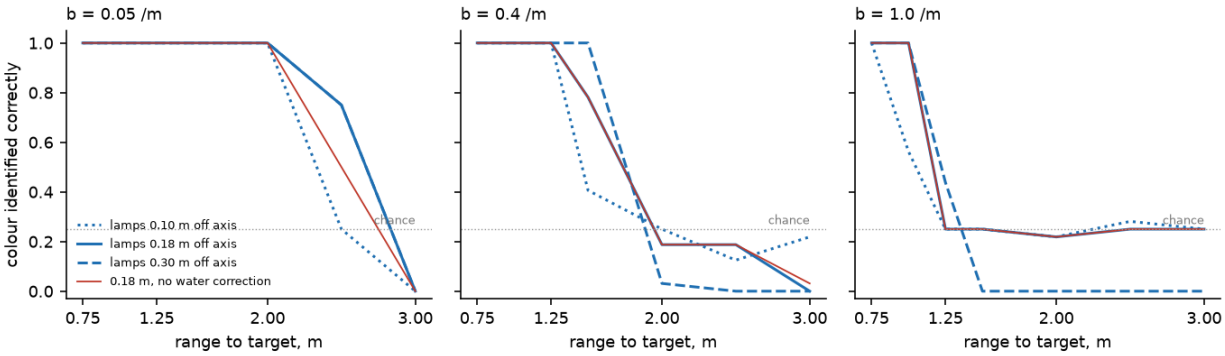


Figure 3: Target colour identified correctly against range, by water clarity and lamp placement, from rendered frames at jittered poses.

6.2 Where the vehicle is on the course, from the image

The camera-alone and shared modes measure the vehicle's position across the course from each frame at 5 Hz, in `vision.CourseEstimator`:

- **gate and obstacles**: posts are found as narrow bright columns. With the distance to the post known, each bearing, rotated by the IMU heading, gives the lateral offset, and an object elsewhere in the frame is rejected by its bearing. One offset has to explain every column at once, so a far post is not paired with a near one. At the gate both posts must be in the frame and each must have its own column; otherwise the frame gives no measurement. Without that rule, once one post had left the edge of the frame close to the gate, the fit took a lone column far beyond it for the missing post and steered the vehicle towards the post it could no longer see; in murky water that was behind every gate contact traced in the random layouts;
- **target**: bearing and range from size of the detection of the target colour;
- **corridor**: a lamp-lit wall dims as the square of its distance, so the log ratio of left to right wall brightness is $s = 2 \ln \frac{w/2+y}{w/2-y}$ for an offset y in a corridor of width w. Inverted, $y = \frac{w}{2} \tanh(gs/4)$ plus a heading term; the gain and heading term are fitted per water clarity on rendered frames from random layouts and tested on held-out frames.

Every estimator is tested on rendered frames called exactly as the closed loop calls it; the gate frames come from random layouts, so what lies beyond the gate is in the test. These held-out errors are also what the shared mode weights the camera by.

RMS lateral error, mm	$b = 0.05$	$b = 0.4$	$b = 1.0$
Corridor, straight section	25	24	19
Corridor, covered passage	62	62	66
Corridor walls, slalom section	63	92	110
Posts, slalom section	20	16	86
Gate posts at 1 m	25	26	25
Target at 1 m	18	15	59

Table 2: Lateral position error of the course estimators on held-out rendered frames. The slalom posts disturb the wall ratio, which is why that section uses the posts. At the gate, a measurement comes back in 83, 75 and 54 % of frames at 1 m.

7 Pilot in the loop

The vehicle is flown by a person looking at the camera feed, so the question is what a pilot with a reaction time, video latency and an imperfect read of the picture achieves, and how much the vehicle's own loops change that. The operator is a crossover model: stick commands are a gain on the perceived error plus a lead term, delayed by 0.25 s of reaction and 0.15 s of video latency, with motor noise on the stick. The pilot's read of the course degrades with range and with the two way attenuation of the vehicle's own light over it.

manual sticks map to thrust; the pilot also flies depth and attitude, which on a positively buoyant vehicle is constant work.

assist depth hold from the pressure sensor and heading and attitude hold from the IMU; vertical speed is taken from the depth signal, not from the simulator. The pilot steers in the horizontal plane only.

camera the assist loops plus the course estimators in the lateral loop, rendered from the camera model every 0.2 s. Open water legs, with nothing to see or hit, are held on heading, and progress along the course is taken as known. On the vehicle the ROS 2 layer runs the corridor part of this mode, which needs only the corridor width; gate and target guidance need the course geometry, which the pilot supplies.

shared the assist loops, with the pilot's lateral command blended with the camera's, each weighted by the inverse of its error variance: the estimator's held-out error for that section and range (Table 2) against the spread of the pilot's read at that range and clarity. In clear water close in, the pilot's read is the better one and the camera's share is small; in murky water at range, the reverse. The camera's share is formed aboard and does not wait for the video delay, and a frame with no measurement leaves the pilot's command alone.

The camera's estimate is differenced at the control rate, so its rate acts in the tick a frame arrives. Smoothing it first with an alpha-beta tracker was tried and added contacts in both camera modes: the estimate does carry frame to frame noise, but the lag the tracker adds cost more than the noise it took out. The ROS 2 pilot node differences the same way. Runs are scored on what an observer records: contacts as events, the gate passed clean or not, the right target touched with the probe or not, and time.

8 Results

8.1 Reference layout

Every run in every row finished, passed the gate clean and touched the right target, so the rows differ in contacts and time. A pilot's contacts are set by the water. In clear and turbid water a manual pilot averages under one contact a run and the assist cuts that further, because the pilot no longer flies depth and attitude. At $b = 1.0$ the pilot's read of the corridor fails with the two way attenuation of the vehicle's own light, and contacts rise to 8.55 manual and 5.85 assisted, with a spread as large as the mean; over the twenty assisted runs, 109 contacts are with the maze walls and 8 with the slalom posts.

The camera alone behaves the other way: 2.50, 2.80 and 2.90 contacts a run at the three clarities, with a small spread, because its error comes from the estimators, which depend little on clarity, and not from a read that fails with range. Its contacts are at the first two slalom posts. The gap beside a post is 72.5 cm, which leaves the 44 cm beam 142 mm a side when centred, and there the wall ratio, which assumes two plain walls, is disturbed by the posts (92 mm RMS at $b = 0.4$ against 24 mm in the straight section), and the post fit is available in 57 % of frames.

The shared mode takes the better of the two wherever it is on the course. Paired seed by seed against the assist, its contacts a run differ by $+0.05 \pm 0.09$ in clear water, -0.05 ± 0.17 in turbid water and -5.40 ± 1.19 in murky water (mean and standard error): no worse where the pilot can see, and in murky water the wall

Mode, water	Finished	Gate clean	Right target	Contacts per run	Time, s
manual, $b = 0.05$	20/20	20	20	0.50 ± 0.59	96.4
manual, $b = 0.4$	20/20	20	20	0.65 ± 0.85	101.9
manual, $b = 1.0$	20/20	20	20	8.55 ± 5.28	134.6
assist, $b = 0.05$	20/20	20	20	0.15 ± 0.36	94.3
assist, $b = 0.4$	20/20	20	20	0.45 ± 0.59	98.9
assist, $b = 1.0$	20/20	20	20	5.85 ± 5.53	129.4
camera, $b = 0.05$	20/20	20	20	2.50 ± 0.67	97.1
camera, $b = 0.4$	20/20	20	20	2.80 ± 0.81	97.2
camera, $b = 1.0$	20/20	20	20	2.90 ± 0.83	107.0
shared, $b = 0.05$	20/20	20	20	0.20 ± 0.40	94.1
shared, $b = 0.4$	20/20	20	20	0.40 ± 0.58	98.5
shared, $b = 1.0$	20/20	20	20	0.45 ± 0.86	128.0

Table 3: Course runs on the reference layout, 20 seeds per row.

contacts all but go (3 over twenty runs). What remains, in every mode that flies the course well, is the first two slalom posts.

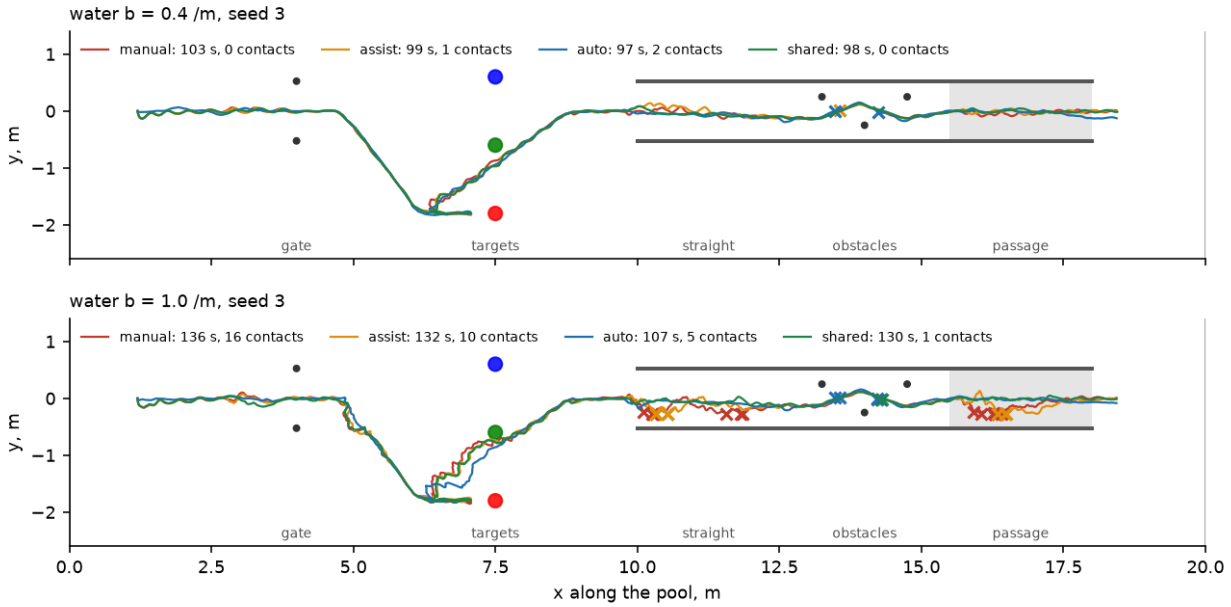


Figure 4: Tracks of one seed in each mode, in turbid and murky water; auto is the camera alone. Crosses mark contacts.

8.2 Random layouts

Mode, water	Finished	Gate clean	Right target	Contacts per run	Time, s
manual, $b = 0.4$	40/40	40	40	1.15 ± 1.01	101.1
manual, $b = 1.0$	40/40	40	40	5.72 ± 3.94	136.8
assist, $b = 0.4$	40/40	40	40	1.27 ± 1.00	98.2
assist, $b = 1.0$	40/40	40	40	4.62 ± 3.34	131.4
camera, $b = 0.4$	40/40	40	40	3.05 ± 0.86	96.6
camera, $b = 1.0$	40/40	40	40	3.48 ± 1.26	108.3
shared, $b = 0.4$	40/40	40	40	1.30 ± 1.05	97.7
shared, $b = 1.0$	40/40	40	40	1.30 ± 1.05	129.6

Table 4: Forty random layouts per row: gate and target line placement, colour order, which colour is the target, corridor width and slalom offsets all vary.

Forty random layouts give the same picture, paired layout by layout. In turbid water the shared mode is level with the assist ($+0.03 \pm 0.14$ contacts a run) and with the manual pilot ($+0.15 \pm 0.17$); in murky water it makes -3.33 ± 0.56 against the assist and -4.42 ± 0.65 against the manual pilot, and the assist's 146 wall contacts fall to 2. The gate was passed clean and the right target touched in every run of every mode; a wrong balloon was touched on the way in a few times (in turbid water 4 manual and 2 shared runs of 40).

The operating rule that follows is to fly shared throughout. It asks the pilot for no decision about when the water is too murky to trust their eyes: the weighting makes that decision from the water clarity, which is a setting, measured from camera frames in the pool before a run.

9 Hardware in the loop

The bench splits the loop the way the vehicle is wired. The host is the world and the surface station: it holds the pool, the true dynamics and the sensor models, runs the pilot, and sends the sensor sample and the pilot packet each tick. The board is the vehicle: it runs the hold loops, the allocation, the governor and the supervisor on what it receives and returns the wrench the thrusters produce. It never sees the course or the truth. The protocol is a framed, CRC-checked link with two message types for the pilot command and the supervisor report; a lost pilot packet is emulated by an acknowledgement in its place, which keeps the bench deterministic.

Over the link, an assisted course finished in 97.68 s against 97.68 s in the pure simulation, with the same contact count (1) and a largest position difference of 2.8×10^{-8} m over 4884 ticks, which is re-allocation round off and nothing else. The safety results are in Section 4. The board and the ESP32 actuator interface have run this protocol on hardware; this bench ran over localhost, and the next step is the same run on the boards.

10 Gazebo model

The vehicle also runs in Gazebo Harmonic under ROS 2 Jazzy, with the same nodes it flies on, against the pool world. Neither the model nor the world is written by hand: `make_model.py` builds the vehicle from the solved budget and the CAD, and `make_world.py` builds the pool from the arena the course runs use, for the reference or any random layout.

- **mass and inertia**: from the budget, about the CG, which is the link origin as it is the origin of the python dynamics; the eight propeller links the thruster plugin needs are taken out of the hull's share, so the totals are the budget's;
- **added mass**: integrated implicitly by the physics engine. The hydrodynamics plugin's own added mass is explicit, and diverges when the heave added mass exceeds the dry mass, as it does here;
- **buoyancy**: Gazebo integrates the volume of the collision shapes, which are also what a post or a wall meets. The contact shapes are the parts that can be met, at their real sizes, the side plates at the thickness that leaves their pocketed volume; one shape that touches nothing carries the rest of the budgeted volume, placed so the total and its centre are the budget's displaced volume and centre of buoyancy;
- **thrusters**: the stations and axes are checked against the allocation matrix when the model is written. The plugin spins each propeller by setting its joint speed; driving it through joint torque, the plugin's default, twists the hull with the reaction;
- **tether**: the course runs' tether drag, applied at the tow point every physics step from inside Gazebo;
- **depth**: from the pose with the depth cell's noise and rate. Gazebo has no water pressure sensor; its air pressure sensor reads the atmosphere.

Case, open loop	Quantity	RMS difference
Rise, zero thrust	depth	0.08 mm
Heave, verticals pushing down	depth	0.04 mm
Surge	position, speed	0.07 mm, 0.34 mm s^{-1}
Surge	pitch	0.09°
Roll, released at 20°	roll	0.18°
Surge with the tether	position, speed	0.09 mm, 0.31 mm s^{-1}

Table 5: The Gazebo model against the python dynamics, same thruster forces from the same start, compared every 20 ms. The python thrusters apply their command at once here, as Gazebo's do.

The two agree to well inside a millimetre in translation; roll differs most, by a fraction of a degree over a swing of 25°. With the model checked, `test_sim.sh` flies the nodes on it: an assisted surge stick moves the

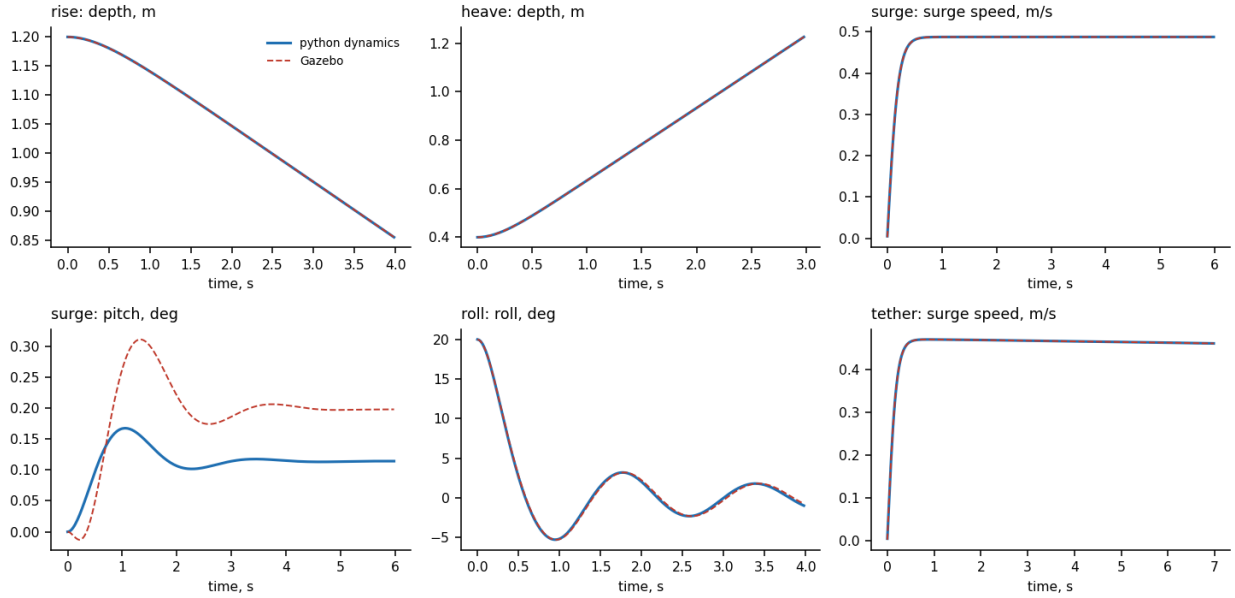


Figure 5: The Gazebo model (dashed) against the python dynamics, open loop.

vehicle forward at its held depth, and silence on the pilot link holds every thruster at zero. Gazebo renders the camera without water, with no absorption or backscatter, so the vision results stay with the camera model of Section 5; the Gazebo camera is for the plumbing.

11 Software

- `simulation/isonavi`: arena, camera, vision, pilot, safety, with the dynamics, sensors and geometry they build on;
- `ros/isonavi_rov_ros`: ROS 2 nodes for the pilot assist, the safety chain, the actuator interface and vision, each a thin wrapper over the module the simulation verified. The pilot node runs the shared mode's blend in the corridor with the same functions the simulation uses, from the worst wall section's error and the pilot's read at 1.5 m, the middle of the operator model's look-ahead along a corridor leg: the camera's share is 0.02, 0.05 and 0.51 in clear, turbid and murky water. Built under Jazzy and run end to end: with the gamepad live the actuator node emits the expected pulse widths, and when it stops the supervisor reports HOLD and every channel goes to neutral;
- `ros/isonavi_rov_description`, `ros/isonavi_rov_gazebo`: the Gazebo model and pool world, their generators, the tether and depth adapters and the check of Section 10;
- `hil`: the bench in Section 9;
- `simulation/vehicle`: budget, CAD and its checks, thruster match, bill of materials.

12 Cost

Prices are in rupees including GST, each line with the basis it rests on (`simulation/vehicle/rov_bom.py`).

13 Verification and what remains

Verified in simulation: the mass, buoyancy and trim budget against the CAD; the thrust envelope with a unit lost; the power path at the governed peak; target identification and the course estimators on rendered frames; the course in four modes across water clarities and random layouts. Verified on the bench: the vehicle-side software reproducing the simulation over the link, and the safety chain under link loss and kill. Verified in Gazebo: the vehicle model against the python dynamics, and the nodes flying it closed loop.

What needs water, in order:

1. a thrust curve for every thruster on the rig before it is fitted, against the 30 N target;
2. a vacuum test of the sealed hull, then a dunk test;
3. trim in a tank, against the solved foam and ballast;

Group	Rupees
Propulsion, eight thrusters	25,472
Hull	8,600
Frame, foam and ballast	7,650
Electronics	33,000
Vehicle	74,722
Tether	8,925
Surface station	31,432
Vehicle, tether and surface	1,15,079
Test equipment and practice pool time	10,400
Contingency	12,548
Total	**1,38,027**

Table 6: Cost by group. 13 of 35 lines are made in house: the thrusters, end caps, frame, buoyancy, lamp housings, kill switch, thermal bridge, tether floats and the thrust rig.

4. camera frames in the pool to calibrate the attenuation coefficient and the corridor gain, which the simulation takes from its model;
5. pilot hours on a mock course, which also measure the pilot's read error that the shared mode weights the camera against.

The hydrodynamic coefficients are scaled from a reference vehicle rather than measured, and the operator model stands in for a real pilot; both are replaced by measurement in the steps above.